

Bootstrap e jQuery

2025/2

FACOM32603 - Desenvolvimento Web I

Prof. Ian R. Cunha
ianresende@ufu.br

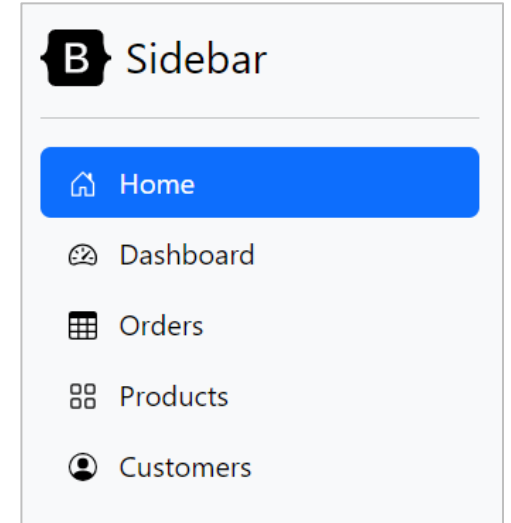
Introdução ao Framework Bootstrap



- É um *framework* front-end muito popular
- Gratuito e de código aberto
- Utiliza HTML, CSS e JavaScript
- Foco em páginas responsivas, mobile-first
- Lançado em 2011 pelo Twitter
- Atualmente, se encontra na versão 5.3
- Documentação: <https://getbootstrap.com/docs>

Introdução ao Framework Bootstrap

- Disponibiliza uma coleção de classes CSS
- Melhoria de layouts, botões, formulários etc.
- Disponibiliza uma coleção de ícones e templates



 Simple header

Home

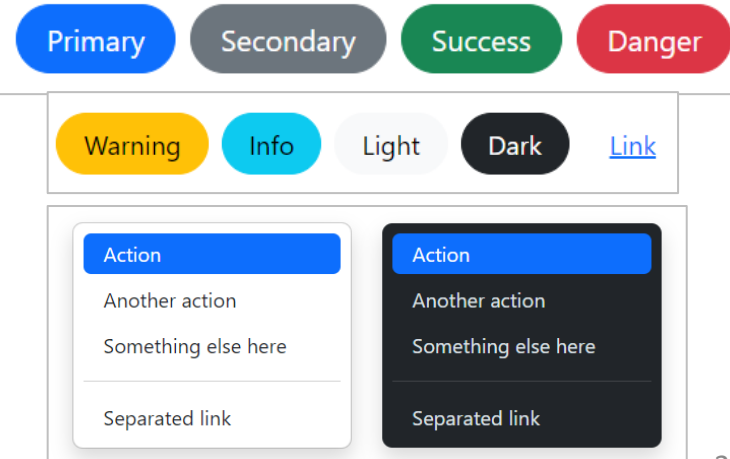
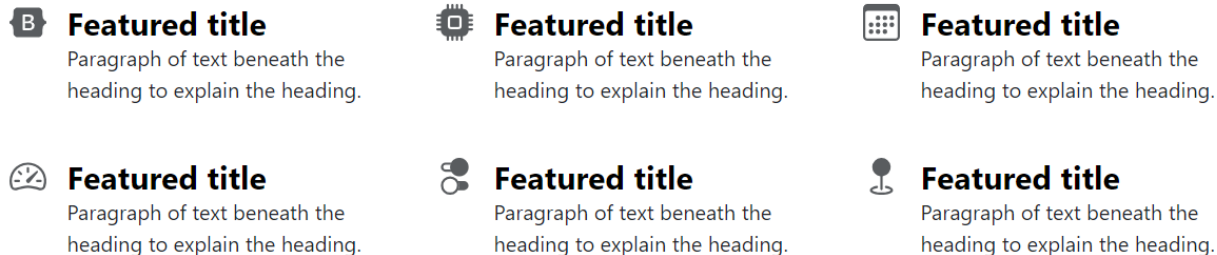
Features

Pricing

FAQs

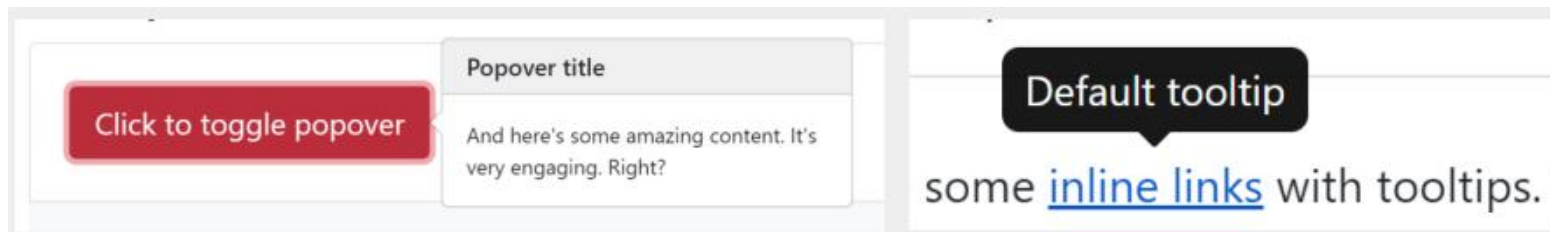
About

Icon grid



Como usar o Bootstrap 5

- Tag de responsividade (`<meta name="viewport" ...>`)
- Arquivo CSS do Bootstrap
- Arquivo JavaScript do Bootstrap
- Arquivo JavaScript da biblioteca Popper (opcional)
 - Caso for utilizar menus suspensos (dropdowns), popovers e tooltips



Nota: antes da versão 5, o Bootstrap também utilizava a biblioteca jQuery.

Como usar

Duas formas de incluir os arquivos:

1. Via CDN

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet"
integrity="sha384-sRl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkulAvNWLzeN8tE5YBujZqJLB" crossorigin="anonymous">

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js" integrity="sha384-
FKyoEForCGlyvwx9Hj09JcYn3nv7wiPVLz7YYwJrWVcXK/BmnVDxM+D2scQbITxI" crossorigin="anonymous"></script>
```

2. Download dos arquivos

- <https://getbootstrap.com>

“Hello, world!” com Bootstrap

- Utilize o código disponível em <https://getbootstrap.com/docs/5.3/getting-started/introduction/>
- Crie um container básico com a classe **container** do Bootstrap
 - Essa classe possibilita que o conteúdo apareça centralizado em telas largas e expandido em telas estreitas
- Inclua 3 parágrafos com qualquer texto
 - Utilize a ferramenta <https://www.lipsum.com/>

Hello, world!

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

There are many variations of passages of Lorem Ipsum available, but the majority have suffered alteration in some form, by

Hello, world!

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

There are many variations of passages of Lorem Ipsum available, but the majority have suffered alteration in some form, by injected humour, or randomised words which don't look even slightly believable. If you are going to use a passage of Lorem Ipsum, you need to be sure there isn't anything embarrassing hidden in the middle of text. All the Lorem Ipsum generators on the Internet tend to repeat predefined chunks as necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with a handful of model sentence structures, to generate Lorem Ipsum which looks reasonable. The generated Lorem Ipsum is therefore always free from repetition, injected humour, or non-characteristic words etc.

“Hello, world!” com Bootstrap

```
<div class="container">  
  <p>Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.</p>  
  <p>Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.</p>  
  <p>Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.</p>  
</div>
```

Hello, world!

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

There are many variations of passages of Lorem Ipsum available, but the majority have suffered alteration in some form, by injected humour, or randomised words which don't look even slightly believable. If you are going to use a passage of Lorem Ipsum, you need to be sure there isn't anything embarrassing hidden in the middle of text. All the Lorem Ipsum generators on the Internet tend to repeat predefined chunks as necessary, making this the first true generator on the Internet. It uses a dictionary of over 200 Latin words, combined with a handful of model sentence structures, to generate Lorem Ipsum which looks reasonable. The generated Lorem Ipsum is therefore always free from repetition, injected humour, or non-characteristic words etc.

Hello, world!

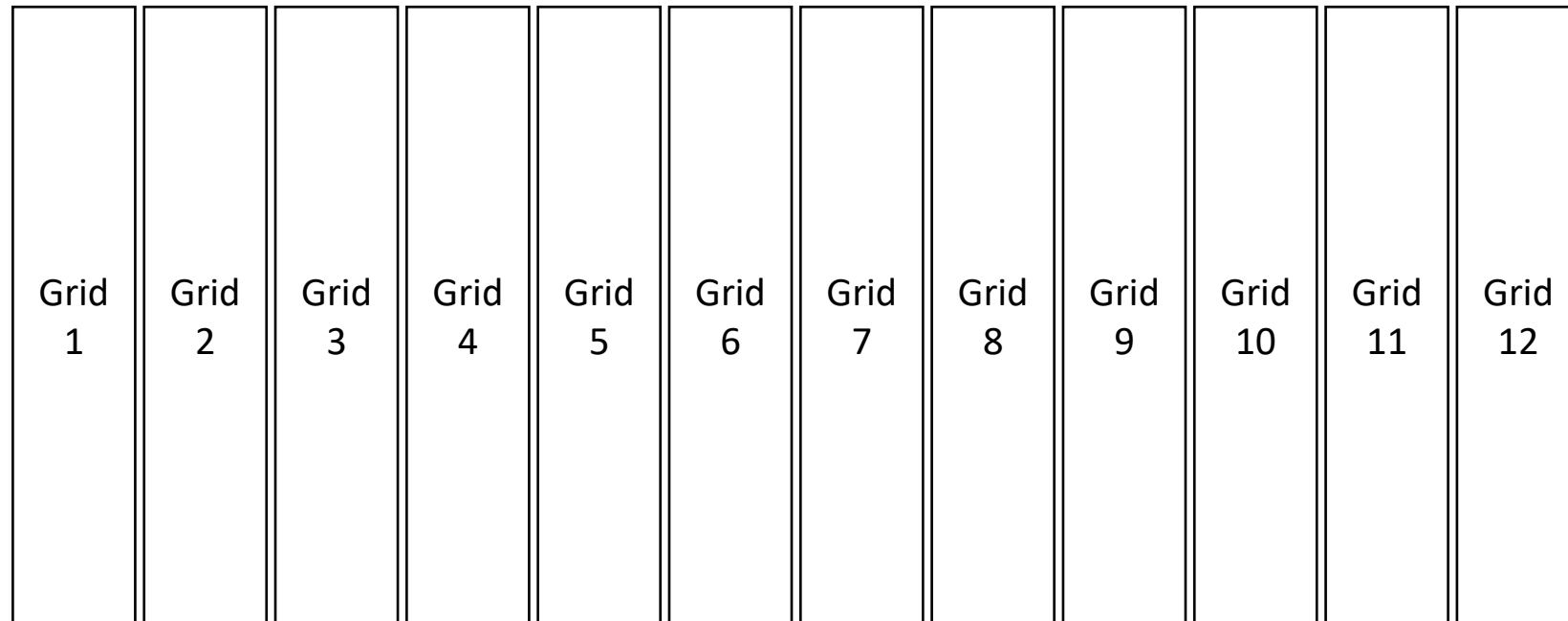
Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

The standard chunk of Lorem Ipsum used since the 1500s is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

There are many variations of passages of Lorem Ipsum available, but the majority have suffered alteration in some form, by

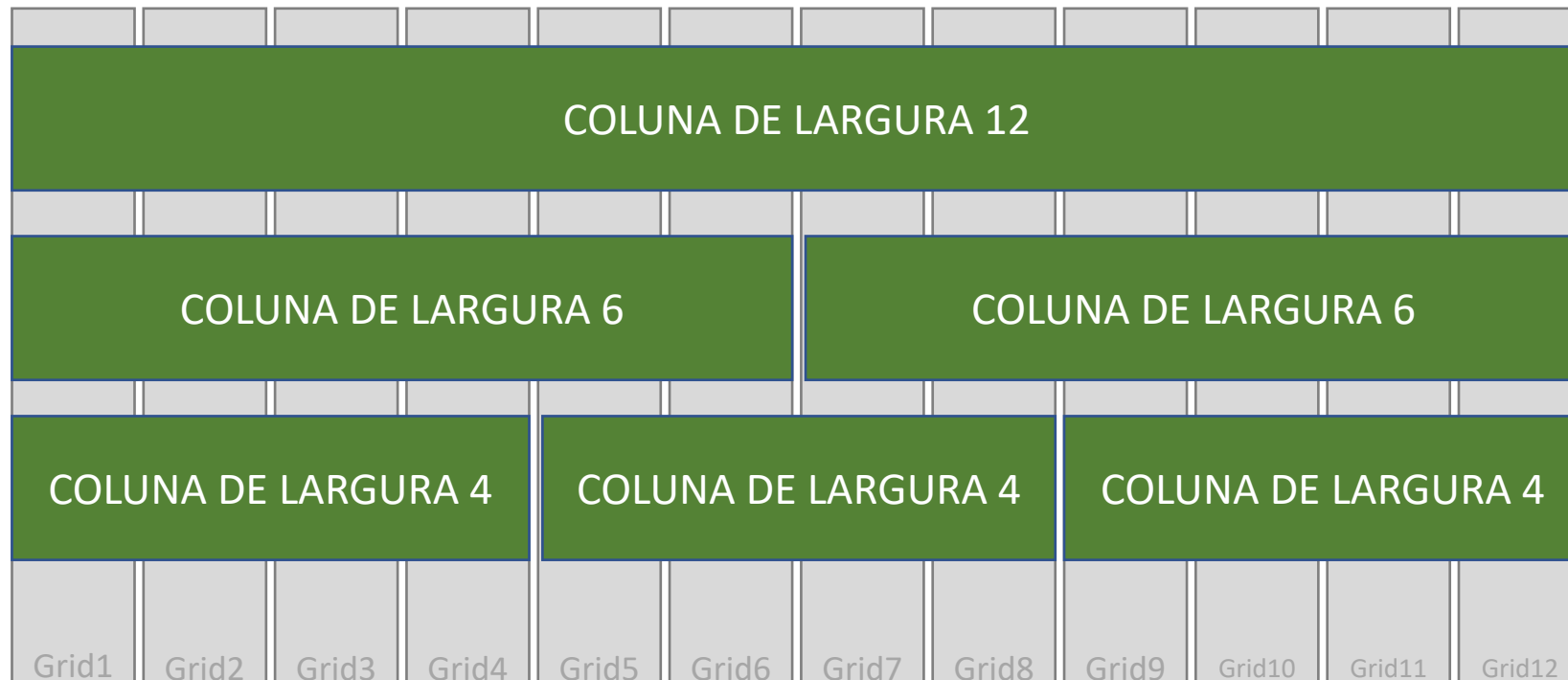
Sistema de grades (grids)

- O Bootstrap possibilita a divisão da página HTML em até 12 colunas de mesma largura
- Baseado no módulo flexbox da CSS e em media queries



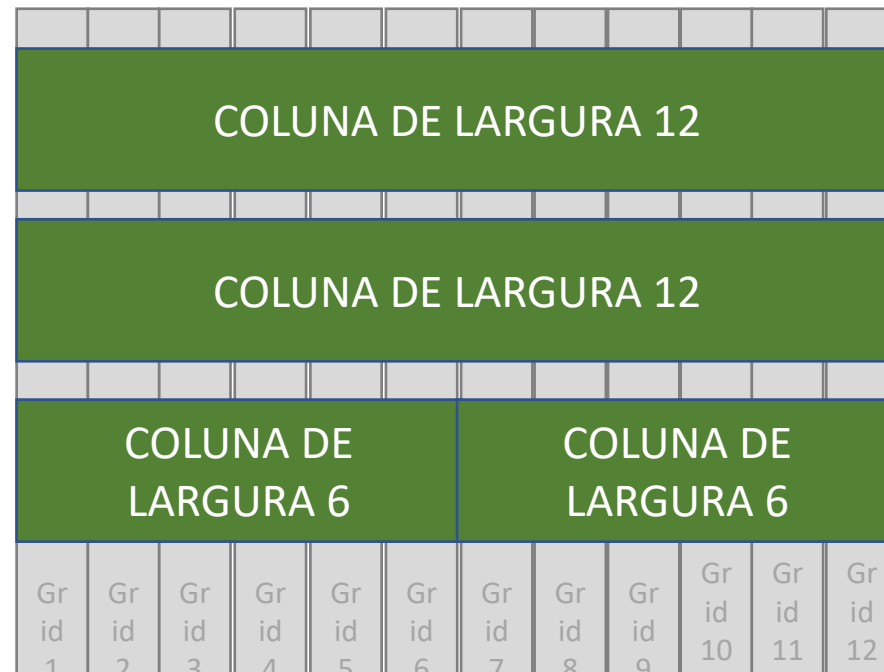
Sistema de grades (grids)

- Com os grids, é possível criar colunas de diferentes tamanhos
- Ideia central: colunas na horizontal para telas grandes e colunas empilhadas na vertical para telas estreitas



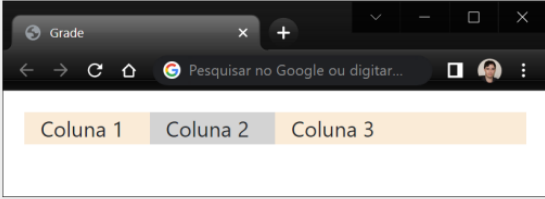
Sistema de grades (grids)

- Com os grids, é possível criar colunas de diferentes tamanhos
- Ideia central: colunas na horizontal para telas grandes e colunas empilhadas na vertical para telas estreitas



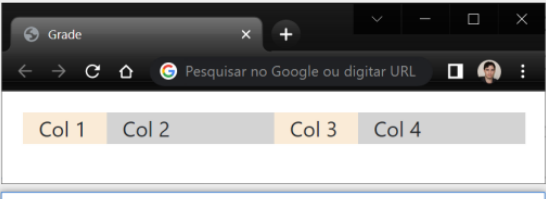
Classes Container - Row - Col

- Classes adicionadas a containers, como o `<div>`
- A classe CSS `container` é um componente necessário para habilitar o sistema de grids
 - `.container` = largura fixa
 - `.container-fluid` = largura total do viewport
- Dentro de um container pode ter várias linhas (classe CSS `row`)
- As colunas (classe CSS `col`) são colocadas dentro de cada linha



```
<div class="row">  
  <div class="col-3">Col 1</div>  
  <div class="col-3">Col 2</div>  
  <div class="col-6">Col 3</div>  
</div>
```

Duas primeiras colunas de mesma largura
Terceira coluna com o dobro das outras



```
<div class="row">  
  <div class="col-2">Col 1</div>  
  <div class="col-4">Col 2</div>  
  <div class="col-2">Col 3</div>  
  <div class="col-4">Col 4</div>  
</div>
```

Colunas 2 e 4 com dobro da largura de 1 e 3

Breakpoints (grid tiers)

`col` { `-breakpoint` } { `-largura` }

opcional opcional

- Definem um limiar para ajuste do layout das colunas
- As colunas aparecem na **horizontal** em viewports com largura **igual ou superior** ao breakpoint
- As colunas aparecem empilhadas na **vertical** em viewports com largura **inferior** ao breakpoint (ponto de quebra)
 - xs (dispositivos muito pequenos): <576px
 - sm (dispositivos pequenos): ≥576px
 - md (dispositivos médios): ≥768px
 - lg (dispositivos grandes): ≥992px
 - xl (dispositivos muito grandes): ≥1200px
 - xxl (dispositivos extremamente grandes): ≥1400px

```
<div class="container">  
  <div class="row">  
    <div class="col-lg-4">Coluna 1</div>  
    <div class="col-lg-4">Coluna 2</div>  
    <div class="col-lg-4">Coluna 3</div>  
  </div>  
</div>
```

Por padrão, a classe `col` sem indicação de breakpoint utiliza a configuração de colunas na horizontal.

Gutters

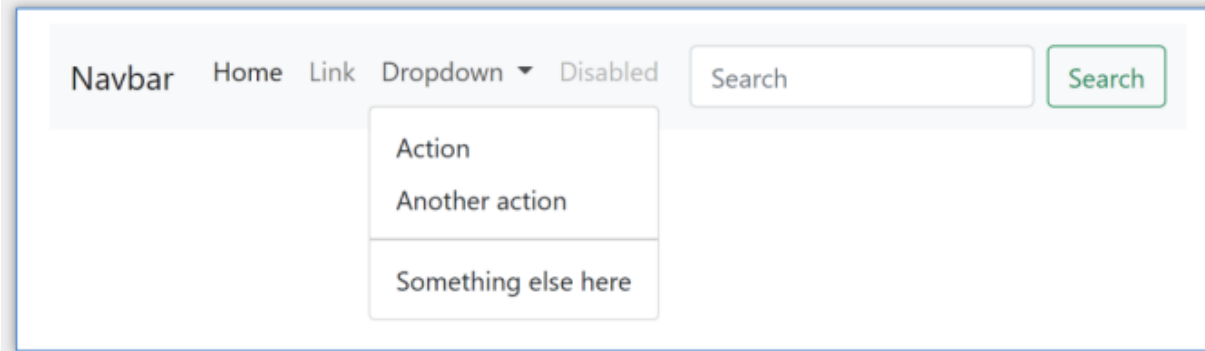
- Espaçamentos na grade entre colunas ou linhas
- Classes CSS no Bootstrap 5
 - `gx-*` → gutter horizontal (espaçamento entre as colunas)
 - `gy-*` → gutter vertical (espaçamento entre as linhas)
 - `g-*` → gutter horizontal e vertical
 - `*` pode ser valores de 0 a 5

Navbar responsiva

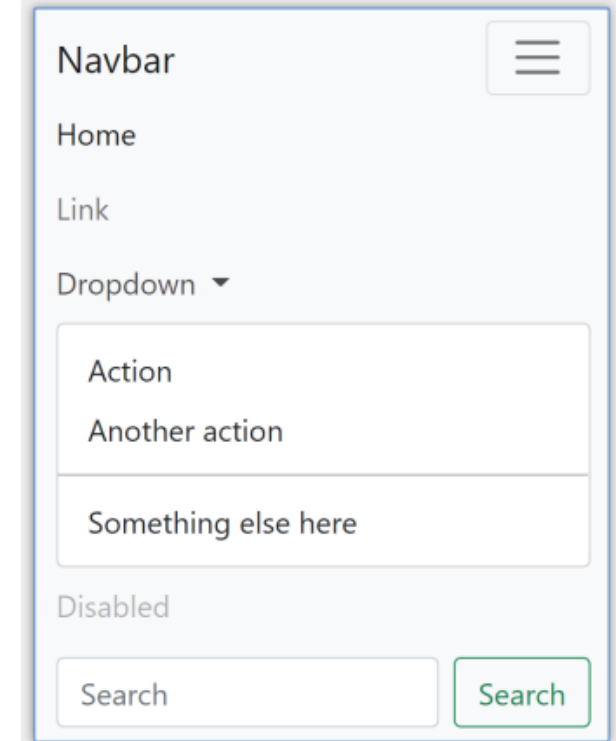
- As barras de navegação e seu conteúdo são `fluid` por padrão

```
<nav class="navbar navbar-expand-sm">  
...  
</nav>
```

Barra de navegação em dispositivos com tela larga



Barra com botão em tela estreita



<https://getbootstrap.com/docs/5.3/components/navbar/>

Botões contextuais

- Classes CSS

.btn

.btn-primary

.btn-secondary

.btn-success

.btn-danger

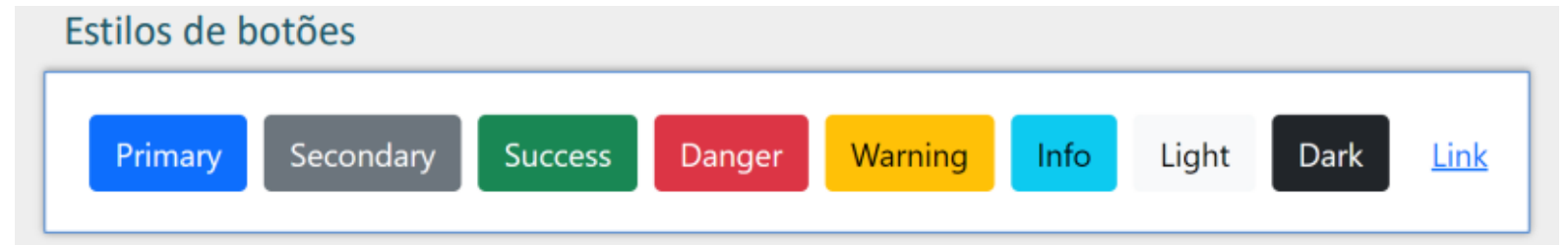
.btn-warning

.btn-info

.btn-light

.btn-dark

.btn-link



```
<button type="button" class="btn btn-primary">Primary</button>
```

Tabelas

```
<table class="table table-striped table-hover">  
...  
</table>
```

Classe Bootstrap	Descrição
.table	Classe principal. Tabela com divisões horizontais
.table-striped	Adiciona cores alternadas às linhas
.table-bordered	Adiciona bordas em todas as células
.table-hover	Adiciona cor à linha quando apontada pelo mouse

#	First	Last	Handle
1	Mark	Otto	@mdo
2	Jacob	Thornton	@fat
3	Larry the Bird		@twitter

Formulários

Classe Bootstrap	Descrição
<code>.form-group</code>	Para qualquer agrupamento de campos de formulário
<code>.form-control</code>	Para campos textuais - input e textarea <code>.form-control-lg</code> : mais largo <code>.form-control-sm</code> : mais fino
<code>.form-label</code>	Para rótulos (label) desses campos
<code>.form-select</code>	Para o elemento <code><select></code>
<code>.form-file</code>	Para campo do tipo file
<code>.form-check</code>	Para grupos de checkboxes e rádios
<code>.form-check-input</code>	Para campos input do tipo checkbox e radio <code>.form-check-label</code> : para rótulos (label) desses campos <code>.form-switch</code> : exibição estilo chave liga/desliga 

Formulários

Formulário com Bootstrap

E-mail

Email

Senha

Senha

Radios

☒ Opção1

☐ Opção2

☐ Opção 3 (desabilitada)

Checkboxes

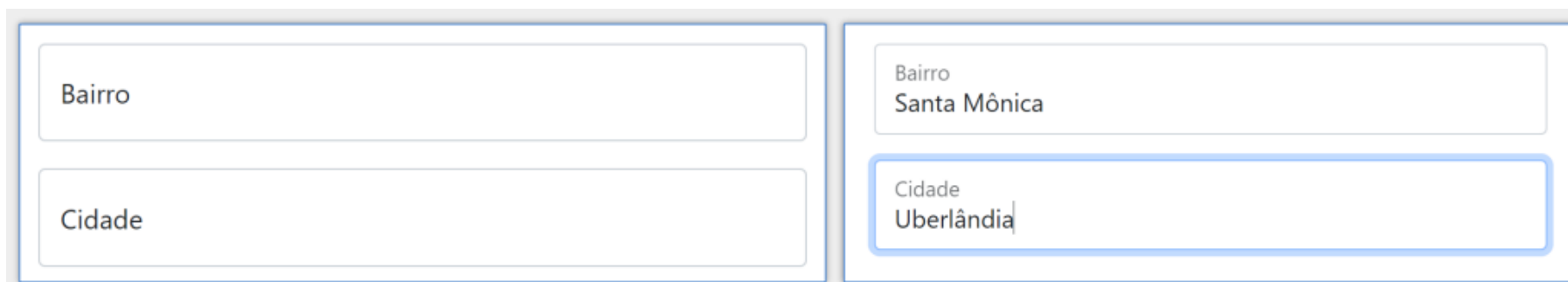
☐ Clique aqui

☐ Clique aqui

Enviar

Campos com Rótulos Flutuantes no Bootstrap 5

- O container do campo/rótulo deve usar a classe `form-floating`
- O label deve ser inserido depois do input
- É obrigatório o atributo `placeholder` com um valor qualquer.
- O conteúdo do label é utilizado como rótulo flutuante (e não o placeholder)



The image displays two side-by-side examples of Bootstrap 5 form-floating components. Each example consists of a container with the class `form-floating` containing two input fields. The left example shows the default state where the labels 'Bairro' and 'Cidade' are floating above empty input fields. The right example shows the state after user input, where the text 'Santa Mônica' and 'Uberlândia' are entered into the respective fields, and the labels 'Bairro' and 'Cidade' are now positioned above the entered text.

Campos com Rótulos Flutuantes no Bootstrap 5

```
<div class="form-floating">  
  <input type="text" class="form-control" id="bairro" placeholder="">  
  <label for="bairro"> Bairro </label>  
</div>  
  
<div class="form-floating">  
  <input type="text" class="form-control" id="cidade" placeholder="">  
  <label for="cidade"> Cidade </label>  
</div>
```

Bairro
Santa Mônica

Cidade
Uberlândia

Alertas

- Classe CSS geral:
 .alert
- Classes CSS
 .alert-primary
 .alert-secondary
 .alert-success
 .alert-danger
 .alert-warning
 .alert-info
 .alert-light
 .alert-dark

```
<div class="alert alert-success" role="alert">  
    Operação realizada com sucesso!  
</div>
```

Success! Indica uma ação realizada com sucesso.

Info! Indica uma informação neutra.

Warning! Indica um aviso que precisa de atenção.

Danger! Indica uma ação potencialmente negativa ou perigosa.

x

Classe .alert-dismissible

Modal

- É uma caixa de diálogo (popup) para realizar interações com o usuário
- As janelas modais são construídas com HTML, CSS e JavaScript
- São posicionadas sobre todo o resto do documento e removem a rolagem do `<body>` para que o conteúdo modal role



Modal

`class="modal"`

`id="meuModal"`

Uso dos seguintes atributos no botão que vai abrir a modal (também é possível via JavaScript):
`data-bs-toggle="modal"`
`data-bs-target="#meuModal"`

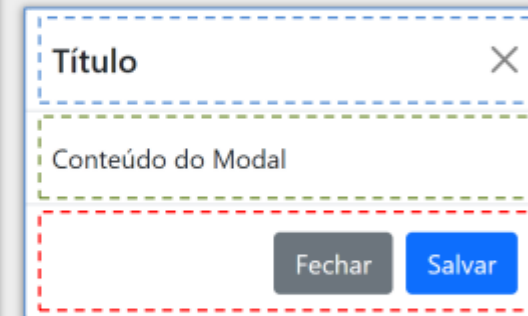
```
<div class="modal fade" id="meuModal">
  <div class="modal-dialog">
    <div class="modal-content">

      <div class="modal-header">
        <h5 class="modal-title">Título</h5>
        <button data-bs-dismiss="modal"></button>
      </div>

      <div class="modal-body">
        Conteúdo do Modal
      </div>

      <div class="modal-footer">
        <button data-bs-dismiss="modal">Fechar</button>
        <button>Salvar</button>
      </div>

    </div>
  </div>
</div>
```



Modal - observações

- Atributos da div principal da janela modal
 - `class="modal"`
 - `id="meuModal"`
- Atributos do botão para abrir a janela modal
 - `data-bs-target="#meuModal"`
 - `data-bs-toggle="modal"`
- Atributos do botão interno para fechar a janela modal
 - `data-bs-dismiss="modal"`

Exercício

- Implemente uma página que contenha os seguintes elementos estilizados com bootstrap:
 - a) Um **menu de navegação** com navbar
 - b) Uma **tabela**
 - c) Um **formulário** com campos input com rótulos flutuantes, checkbox, radiobutton, caixa de seleção (dropdown) e um botão
 - d) Uma div de mensagens de alerta
- No clique do botão, deve abrir uma janela modal.

Referências

- <https://getbootstrap.com/docs>
- <https://icons.getbootstrap.com>
- <https://fonts.google.com/icons>
- <https://wiki.lib.sun.ac.za/images/0/07/Bootstrap-tutorial.pdf>